
Noise thresholds for adaptive variational quantum estimation of many-body Berry phases

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Abstract

Adaptive variational quantum dynamics simulation (AVQDS) computes the Berry phase of an interacting fermion chain with circuits of only a few hundred entangling gates—orders of magnitude shallower than Trotterized adiabatic evolution—which makes it a leading candidate for measuring topological invariants on near-term hardware. All published evidence for the method, however, is noiseless: its celebrated robustness is robustness to *non-adiabaticity* under exact arithmetic, not to stochastic device errors. Here we quantify how much hardware noise the method actually tolerates. We re-implement the full pipeline (adaptive ground-state preparation, McLachlan dynamics along a time-reversed adiabatic loop, ancilla interferometry), reproduce the noiseless benchmarks for the dimerized Fermi–Hubbard chain at interaction strengths $U = 0$ and $U = 10$, and then subject *every measured quantity* the method needs—the interferometric overlap *and* the energy and geometric-connection integrands of its classical phase reconstruction—to a trapped-ion-like error model (two-qubit depolarizing noise, T_1/T_2 relaxation, readout errors, finite sampling) in exact density-matrix simulation. Using a pre-registered accuracy criterion, we extract noise thresholds in device units: with 4096 shots per circuit the topological phase remains correctly resolved below an equivalent two-qubit error of $\approx 8.4 \times 10^{-4}$ for the noninteracting circuits but only $\approx 1.3 \times 10^{-5}$ at $U = 10$ —a $64.2\times$ gap far exceeding their $\sim 5\times$ circuit-size ratio—with an incoherent-noise-free shot floor between ~ 16 and ~ 234 shots per circuit. Decomposing the error explains both numbers: the interferometric phase is remarkably bias-immune (noise shrinks both measured quadratures symmetrically, leaving the phase intact until the contrast meets the shot noise), and the method fails instead through its *dynamical-phase leg*, where depth-asymmetric attenuation of the measured energy between the forward and return half-loops breaks the time-reversal cancellation and is amplified by the loop time and by the energy scale—the lever arm that makes the interacting system disproportionately fragile, and that makes the reconstructed phase *more* fragile than the quantum state carrying it. The same accounting yields a nuanced verdict on circuit compression: at equal shots per circuit the adaptive method tolerates only $\approx 2.2\times$ more noise than a $30\times$ deeper matched-accuracy Trotter interferometer—whose all-interferometric readout is bias-free—while readout calibration plus zero-noise extrapolation, which attacks exactly the dynamical-phase bias, widens the adaptive method’s budget by $4.1\times$ at $U = 10$. Our results give device-facing targets for measuring interacting topological invariants with adaptive variational algorithms, and identify the classical dynamical-phase reconstruction—not the interferometry—as the component that must be defended against noise.

1 Introduction

The Berry phase [1] is the elementary geometric object behind band topology, the quantum Hall effect, and topological insulators and superconductors [2, 3]. For a one-dimensional crystalline system with time-reversal symmetry it is quantized: $\varphi_B = 0$ in a trivial phase and $\varphi_B = \pi$ in a topologically nontrivial one, so a single phase measurement classifies the phase of matter. Computing φ_B for *interacting* systems is classically expensive, which has motivated a family of quantum algorithms that evolve a register around a closed parameter loop and read the accumulated geometric phase off an ancilla [4, 5]. The bottleneck of the direct approach is circuit depth: a Trotterized adiabatic loop for even a four-site Fermi–Hubbard chain requires 10^5 – 10^6 entangling gates [6].

Mootz and Yao [6] recently showed that this depth can be compressed by three orders of magnitude. Their method prepares the ground state with the adaptive variational quantum imaginary-time algorithm (AVQITE) [7], transports it around a time-reversed adiabatic loop with adaptive variational quantum dynamics simulation (AVQDS) [8, 9], and reconstructs the Berry phase as the sum of an interferometrically measured overlap phase and a classically integrated global phase. On the dimerized Fermi–Hubbard (Su–Schrieffer–Heeger–Hubbard, SSHH) chain [10, 11] the method reproduces exact-diagonalization Berry phases to 10^{-3} with peak depths of 47–279 layers, and it does so even at ramp speeds that leave the state far from the instantaneous ground state—a striking noiseless error cancellation between its two phase contributions.

What none of this establishes is whether the method survives *hardware* noise. The published results are exact state-vector simulations, and the two robustness statements are logically independent: the non-adiabatic cancellation is a property of two phase legs that co-evolve deterministically along the same variational trajectory, whereas gate errors, relaxation, readout errors, and shot noise strike the *measured* ingredients of those legs independently and stochastically. Because the final observable is a phase built from a measured probability, finite sampling is irreducible even on fault-tolerant hardware; and because one of the phase legs is a time integral of the measured energy, noise on that leg is amplified by the loop duration. Whether the shallow-circuit advantage of the adaptive method translates into a usable noise budget on a real device is therefore a quantitative question, and it is the question this paper answers.

Contributions. (i) We re-implement the complete AVQITE+AVQDS Berry-phase pipeline from scratch, keeping every algorithmic element of Ref. [6] fixed, and reproduce its noiseless anchors on the SSHH chain in both interaction regimes and both topological phases, cross-validated against the authors’ published data records (Sec. 2–4). (ii) We define a *whole-pipeline* noise study: a trapped-ion-referenced error model acts, in exact density-matrix simulation, on every circuit the method must execute—the ancilla interferometry *and* the energy and geometric-connection measurements that feed the global phase—with all outcomes drawn by genuine finite sampling (Sec. 3). (iii) Under a pre-registered tolerance and extraction rule we report noise thresholds with uncertainties, in device units, for all four configurations, together with the irreducible shot floor (Sec. 5). (iv) We dissect the failure mechanism—which phase leg breaks, through which channel, and why—and give a quantitative model of the dominant error: a depth-asymmetric, loop-time-amplified drift of the dynamical-phase leg (Sec. 6). (v) We test the folklore that circuit compression translates into noise tolerance against a matched-accuracy Trotter baseline—finding a much smaller dividend than the $30\times$ gate ratio suggests, for a structural reason—and demonstrate that standard, low-overhead mitigation (calibrated readout inversion plus zero-noise extrapolation), which attacks precisely the failing leg, widens the threshold by 2 – $4\times$ (Secs. 7–8).

2 The algorithm under study

We characterize the algorithm of Ref. [6] exactly as published; this section fixes notation and records the implementation details a reader needs to reproduce our study.

2.1 Model and reference invariant

The SSHH chain with N sites, staggered hopping and on-site repulsion is

$$\hat{\mathcal{H}} = \sum_{j,\sigma} \left[t_{j,j+1} \hat{c}_{j+1,\sigma}^\dagger \hat{c}_{j,\sigma} + \text{h.c.} \right] + U \sum_j \hat{n}_{j,\uparrow} \hat{n}_{j,\downarrow} - \frac{U}{2} \sum_{j,\sigma} \hat{n}_{j,\sigma}, \quad t_{j,j+1} = t (1 + (-1)^j \delta), \quad (1)$$

with periodic boundaries. We use $N = 4$, $t = 1$, and both $U = 0$ and $U = 10$. A twist $\hat{c}_{j+N,\uparrow} = e^{i\lambda} \hat{c}_{j,\uparrow}$ applied to the spin-up species only threads the ring, and the Berry phase of the many-body ground state over the loop $\lambda : 0 \rightarrow 2\pi$ is the topological invariant: $\varphi_B = \pi$ for $\delta < 0$ (twisted bond strong) and $\varphi_B = 0$ for $\delta > 0$, with the transition at $\delta = 0$ [6, 11]. Jordan–Wigner mapping [12] with the two spin species interleaved yields an 8-qubit Hamiltonian $\hat{\mathcal{H}}(\lambda) = \hat{H}_0 + \cos \lambda \hat{H}_c + \sin \lambda \hat{H}_s$ containing 18 ($U = 0$) or 22 ($U = 10$) distinct Pauli strings. Our construction is validated against an independent second-quantized diagonalization in the occupation basis (agreement at machine precision for all λ) and is spectrum-identical to the Hamiltonian records shipped with Ref. [6]’s data. The reference invariant is computed at runtime from a discretized Wilson loop [13, 14] over 400 points, $\varphi_B^{\text{ED}} = -\text{Im} \ln \prod_k \langle G(\lambda_k) | G(\lambda_{k+1}) \rangle$.

2.2 Adaptive state preparation and transport

Both adaptive components represent the state as $|\Psi[\theta]\rangle = \prod_{\mu=1}^{N_\theta} e^{-i\theta_\mu \hat{A}_\mu} |\Psi_0\rangle$ with Pauli-string generators $\hat{A}_\mu$ appended on demand. Time evolution follows McLachlan’s variational principle [15, 16]: with $M_{\mu\nu} = 2 \text{Re}[\langle \partial_\mu \Psi | \partial_\nu \Psi \rangle + \langle \partial_\mu \Psi | \Psi \rangle \langle \partial_\nu \Psi | \Psi \rangle]$ and $V_\mu = 2 \text{Im}[\langle \partial_\mu \Psi | \hat{\mathcal{H}} | \Psi \rangle + \langle \Psi | \partial_\mu \Psi \rangle \langle \hat{\mathcal{H}} \rangle]$, the parameters obey $\sum_\nu M_{\mu\nu} \dot{\theta}_\nu = V_\mu$, and the residual McLachlan distance $L^2 = 2 \text{var}[\hat{\mathcal{H}}] - \sum_\mu V_\mu \dot{\theta}_\mu$ measures the deviation from exact evolution. Whenever $L^2 \geq L_{\text{cut}}^2$ the ansatz is expanded TETRIS-style [9]: candidate generators are ranked by the L^2 they would achieve, the best is appended, and further disjoint-support candidates are added in rank order until the register is covered, all with $\theta = 0$. We hold the published hyperparameters fixed: $L_{\text{cut}}^2 = 10^{-4}$ for the loop dynamics, adaptive fourth-order Runge–Kutta steps with $\max_\mu |\dot{\theta}_\mu \delta t| < \delta\theta_{\text{max}} = 0.01$, the Hamiltonian-string operator pool for AVQDS, and the 616-element odd- $\hat{\sigma}^y$ singles/doubles pool for AVQITE [7]. For AVQITE we use $L_{\text{cut}}^2 = 10^{-2}$, the value used for the authors’ shipped ground-state ansatz records, which reproduces their preparation circuits (20 CNOTs at $U = 0$, 68 CNOTs at $U = 10$ vs. their 92) and preparation infidelities (2.1×10^{-11} and 4.7×10^{-4} , vs. their 2×10^{-8} and 4.7×10^{-4}). Two implementation choices required by reproducibility are disclosed here: (i) we solve the McLachlan system with Tikhonov regularization $(M + \varepsilon \mathbb{K}) \dot{\theta} = V$, $\varepsilon = 10^{-7}$, a standard stabilization of variational dynamics [16, 27] that leaves every anchor observable unchanged at the 10^{-4} level but makes $\dot{\theta}(t)$ a continuous function of time (a minimal-norm pseudo-inverse jumps between near-degenerate directions, which would make the global-phase integrand unmeasurable at finite sampling cadence); (ii) the reference product state $|\Psi_0\rangle$ is the half-filled state used by the authors’ records: spin-up on sites 1, 2 / spin-down on sites 3, 4 for $\delta < 0$, and both species on sites 1 and 4 for $\delta > 0$ (the imaginary-time flow from the former stalls on an excited stationary state for $\delta > 0$; the authors’ archive contains the same reference-state switch).

2.3 Loop protocol and phase reconstruction

The Berry phase is accumulated on a time-reversed adiabatic loop of total wall-clock duration T : the twist advances as $\lambda(s) = 2\pi s/T$ throughout, the first half ($s < T/2$, $\lambda : 0 \rightarrow \pi$) evolves forward in time, and the second half ($\lambda : \pi \rightarrow 2\pi$) applies time-reversed propagation, $i d|\Psi\rangle/ds = -\hat{\mathcal{H}}|\Psi\rangle$. For the time-reversal-invariant Hamiltonian (1) the dynamical phase cancels between the halves while the geometric phase adds. Because the variational ansatz carries no global phase, the method reconstructs φ_B from three pieces:

$$\varphi_B = \varphi_{\text{qc}} + \varphi_{G,1} + \varphi_{G,2}, \quad \varphi_{G,1} = -\int_0^T ds \sigma(s) \langle \hat{\mathcal{H}} \rangle_{\theta(s)}, \quad \varphi_{G,2} = \int_0^T ds \sum_\mu \text{Im} \langle \partial_\mu \Psi | \Psi \rangle \dot{\theta}_\mu, \quad (2)$$

where $\sigma(s) = \pm 1$ is the leg sign and $\varphi_{\text{qc}} = \arg \langle \Psi[\theta(0)] | \Psi[\theta(T)] \rangle$ is measured by an ancilla Hadamard test over the controlled circuit $W = U_G^\dagger[\theta(0)] U[\theta(T)]$ [17]. The integrands of $\varphi_{G,1}$ and

$\varphi_{G,2}$ are device observables—the energy $\langle \hat{\mathcal{H}} \rangle$ and the geometric connections $\text{Im} \langle \partial_\mu \Psi | \Psi \rangle = \langle \hat{\mathcal{A}}_\mu \rangle_{\psi_\mu}$, the latter a plain Pauli expectation in the *prefix* state $|\psi_\mu\rangle$ obtained by running the circuit up to gate μ . This anatomy is the pivot of the whole paper: $\varphi_{G,1}$ is a *dynamical-phase* leg whose integrand is an extensive energy, $\varphi_{G,2}$ is a geometric leg, and φ_{qc} is a single interferometric phase; noiseless non-adiabatic errors cancel between φ_{qc} and $\varphi_{G,2}$ [6], and Sec. 6 asks which of these cancellations survive stochastic noise. We verified our global-phase bookkeeping directly: propagating $e^{i\varphi_G} |\Psi[\theta]\rangle$ forward reproduces the exactly evolved state including its phase to 4×10^{-8} rad.

Conventions used throughout: $\varphi_{G,1}$ and $\varphi_{G,2}$ are accumulated continuously (unwrapped) along the loop; $\varphi_{qc} \in (-\pi, \pi]$; the sum is folded once to the principal branch $(-\pi, \pi]$; and every deviation is a *circular* distance, $|\Delta\varphi_B| = |\text{wrap}(\hat{\varphi}_B - \varphi_B^{\text{ED}})| \in [0, \pi]$, so estimates straddling the $\pm\pi$ branch point (where the $U = 10$ raw sums land, e.g. $-9.425 = -3\pi$ before folding) are compared correctly.

3 Noise model and estimation protocol

This section fixes, before any results, the device model, the measurement protocol for every phase leg, the estimator, and the criterion by which a threshold is extracted. The scope of the study is the *measurement* resilience of the method: the ansatz and the variational trajectory $\theta(t)$ are frozen at their noiseless values, and noise strikes every circuit executed to *evaluate* φ_B —the Hadamard tests and all $\varphi_{G,1}/\varphi_{G,2}$ integrand measurements along the frozen trajectory. Feeding sampling noise back into the McLachlan construction itself (a stochastic trajectory) is a distinct, harder question that we leave open (Sec. 11).

3.1 Device model

All circuits are compiled to $\{X, H, S, S^\dagger, R_z, \text{CNOT}\}$ with all-to-all connectivity, the setting of the original resource counts (trapped-ion hardware [18]): a weight- w Pauli rotation costs $2(w-1)$ CNOTs via basis changes and a CNOT ladder, and each *controlled* rotation of the Hadamard test adds one controlled- R_z gadget (2 CNOTs), so the interferometric circuit costs only 2 extra CNOTs per variational parameter, as in Ref. [6]. The base noise profile is trapped-ion-like:

2q depolarizing	1q depolarizing	readout flip	T_1 / T_2	gate times
$p_{2q}^{(0)} = 3 \times 10^{-3}$	$p_{1q}^{(0)} = 10^{-4}$	$\epsilon_{ro}^{(0)} = 3 \times 10^{-3}$	10 s / 1 s	$t_{1q} = 10 \mu\text{s}, t_{2q} = 200 \mu\text{s}$

A single scalar s multiplies all error *rates* ($p_{2q}, p_{1q}, \epsilon_{ro}, 1/T_1, 1/T_2$) against this profile, so the four channels co-vary in the fixed ratio the profile sets; $s = 1$ is the base device and results are quoted both in s and in the equivalent two-qubit error $s p_{2q}^{(0)}$. Channels act on the compiled circuit in exact density-matrix simulation: a depolarizing channel after every CNOT (probability $s p_{2q}^{(0)}$, uniform over the 15 two-qubit Paulis) and every one-qubit gate; amplitude damping plus pure dephasing applied to *all* qubits for the duration of each ASAP-scheduled circuit layer (idle qubits decohere; layer duration t_{2q} if it contains a CNOT, else t_{1q}); and an independent symmetric bit-flip $s \epsilon_{ro}^{(0)}$ on every measured qubit. Shot noise is never emulated: every reported estimate draws binomial/multinomial samples from the exact noisy outcome distributions. The simulator’s channels are unit-tested against analytic expectations (depolarizing contrast factors, e^{-t/T_1} populations, e^{-t/T_2} coherences, confusion matrices) and its gates against dense matrix references.

3.2 Whole-pipeline measurement protocol

At each of 482 quadrature nodes along the frozen trajectory we execute, under the full noise model: (i) the state-preparation circuit truncated at every prefix μ , measuring the connection $\langle \hat{\mathcal{A}}_\mu \rangle$ by basis rotation and readout of its support (one prefix per variational parameter present at that node); and (ii) the full circuit at that node, measuring $\langle \hat{\mathcal{H}} \rangle$ in qubit-wise-commuting groups (10–13 per configuration; the greedy grouping depends on the coefficient ordering; the identity offset is a classical constant). At the loop endpoint we execute the two 9-qubit Hadamard-test circuits (ancilla $|0\rangle$ probability $P_0 = (1 + \text{Re} \langle W \rangle)/2$, and with an extra $S^\dagger$, $(1 + \text{Im} \langle W \rangle)/2$), each carrying 198 ($U = 0$) / 1074 ($U = 10$) CNOTs including the state preparation, the controlled gadgets, and the inverse preparation.

Table 1: Noiseless anchor: reproduction of the adaptive Berry-phase pipeline at $T = 20$ (McLachlan cutoff 10^{-4} , $\delta\theta_{\max} = 0.01$; topological $= \delta = -0.5$ ($U=0$) / -0.6 ($U=10$), trivial $= +0.5$ / $+0.6$). φ_B is the folded Berry phase of our runs; the deviation is the circular distance to the runtime exact-diagonalization target; N_θ /CNOT/depth are end-of-loop ansatz resources from the logged operator selections (parenthetical CNOT and infidelity values: the corresponding archived runs of Ref. [6] at the same (δ, T)). AVQITE columns give the ground-state preparation cost and infidelity.

Configuration	φ_B^{ED}	φ_B	$ \Delta\varphi_B $	N_{CX} (arch.)	depth	$N_{\text{CX}}^{\text{prep}}$	$1-f_{\text{prep}}$	$\max(1-f)$ (arch.)
$U=0$ topological	π	-3.1416	3.5×10^{-5}	120 (352)	15	20	2.1×10^{-11}	1.7×10^{-2} (1.7×10^{-2})
$U=0$ trivial	0	-0.0041	4.1×10^{-3}	124 (164)	12	20	2.1×10^{-11}	2.3×10^{-3} (2.4×10^{-3})
$U=10$ topological	π	-3.1415	7.3×10^{-5}	592 (780)	61	68	4.7×10^{-4}	3.3×10^{-1} (3.3×10^{-1})
$U=10$ trivial	0	-0.0000	1.7×10^{-5}	592 (944)	61	68	4.7×10^{-4}	5.9×10^{-4} (6.5×10^{-4})

Estimator. The overlap phase uses both quadratures, $\hat{\varphi}_{\text{qc}} = \text{atan2}(\widehat{\text{Im}}, \widehat{\text{Re}})$, which keeps the estimator well-conditioned wherever φ_{qc} sits—important because at quantized targets the single-quadrature readout $P_0 = (1 + \cos \varphi)/2$ has vanishing slope, and in our configurations φ_{qc} lands near π (e.g. 1.330 at $U = 0$, $\delta = -0.5$). The phase legs use the frozen trajectory as classical knowledge: with measured node values $\hat{E}_j$ and $\hat{a}_{j\mu} = \langle \hat{A}_\mu \rangle$,

$$\hat{\varphi}_{G,1} = \sum_j W_j^{(1)} \hat{E}_j, \quad \hat{\varphi}_{G,2} = \sum_{j\mu} W_{j\mu}^{(2)} \hat{a}_{j\mu}, \quad (3)$$

where the weights $W_j^{(1)} = \int \text{hat}_j(s) (-\sigma(s)) ds$ and $W_{j\mu}^{(2)} = \int \text{hat}_j(s) \dot{\theta}_\mu(s) ds$ interpolate the measurements linearly between nodes and integrate them against the *exact* stored parameter increments (the velocities oscillate on the integrator timescale; the smooth objects are the state-dependent observables, so this splitting is what makes a coarse measurement cadence viable on hardware). The zero-noise, infinite-shot limit of the estimator reproduces the anchor Berry phases to within 0.0641 rad in the worst configuration ($U=10$ trivial; the other three are below 9.1×10^{-3} ; all logged), a quadrature bias of 1–16% of the tolerance below, included in every reported deviation and never corrected for. Each circuit (each prefix, each energy group, each quadrature) receives S shots; the reference setting is $S_{\text{ref}} = 4096$, for a total sampling budget of $\sim 5.8 \times 10^7$ ($U = 0$) to $\sim 2.8 \times 10^8$ ($U = 10$) shots per Berry-phase estimate.

3.3 Accuracy criterion and threshold rule (pre-registered)

The tolerance is $\epsilon_{\text{tol}} = \pi/8 \approx 0.39$ rad of circular deviation from the exact-diagonalization target (0 or π): four times smaller than the $\pi/2$ decision boundary between the two quantized values, and two orders of magnitude above the noiseless anchor accuracy. For each configuration and noise scale we run 16 independent sampling seeds at S_{ref} and form $|\Delta\varphi_B|$, the seed-mean circular deviation. The threshold s^* is the crossing $|\Delta\varphi_B|(s) = \epsilon_{\text{tol}}$ of the running-maximum envelope over the swept grid $s \in \{0.003, 0.01, 0.02, 0.03, 0.05, 0.1, 0.2, 0.3, 0.5, 1\}$, log-linearly interpolated between the bracketing grid points; uncertainties are 16/84 bootstrap percentiles over seeds (2000 resamples). After the first pass located the $U=10$ crossings inside the widest grid interval, two refinement levels ($s = 0.005, 0.007$) were added—swept for all four configurations—to tighten the interpolation bracket: a disclosed, rule-preserving refinement; every swept point is reported (Table 3). The shot floor S^* applies the same rule on the shots axis $S \in \{64, \dots, 65536\}$ at $s = 0$ (24 seeds), where the deviation decreases with S . The per-interaction threshold quoted for each U is the smaller of its two regimes, i.e. the level below which the trivial and topological phases are *both* rendered accurately. The rule, tolerance, grids, and seed counts were fixed before the sweeps were run and are applied identically to every configuration; all swept points are reported.

4 Noiseless anchor

Table 1 summarizes the reproduction of the noiseless baseline that every noisy number is referenced to. All four configurations reproduce the quantized invariant—to 3.5×10^{-5} , 4.1×10^{-3} , 7.3×10^{-5} , and 1.7×10^{-5} rad respectively—against the runtime Wilson-loop reference, resolving the topological transition. Resources land in the published band: the $U = 10$ circuits are $\sim 5\times$ deeper than $U = 0$, the interacting preparation infidelity (4.7×10^{-4}) matches the published 4.7×10^{-4} almost exactly,

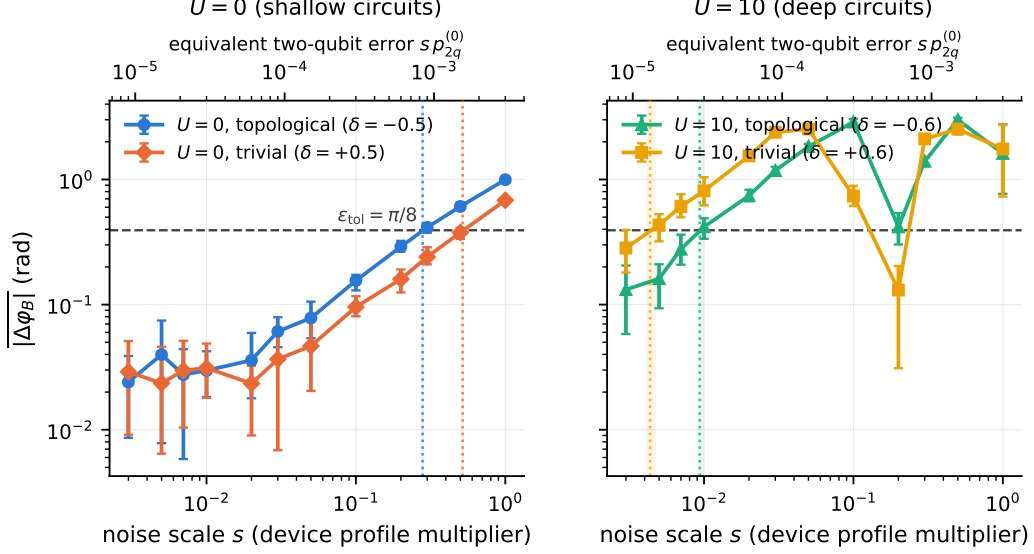


Figure 1: Berry-phase error versus noise strength. Seed-mean circular deviation $|\overline{\Delta\varphi_B}|$ at $S_{\text{ref}} = 4096$ shots per circuit versus the noise scale s (bottom axis) and the equivalent two-qubit error (top axis), for the $U = 0$ (left) and $U = 10$ (right) configurations; error bars are 16/84 seed percentiles over 16 seeds. The dashed line is the pre-registered tolerance $\epsilon_{\text{tol}} = \pi/8$; dotted verticals and shaded bands are the extracted thresholds s^* with bootstrap uncertainties. The deep interacting circuits fail at a $\sim 64.2\times$ smaller noise scale than the shallow noninteracting ones.

and the loop-level diagnostics agree with the authors’ data records: for $U = 10$, $\delta = -0.6$, $T = 20$ our maximum loop infidelity is 0.334 vs. the archived 0.334, and our dynamical-phase residual $\varphi_{G,1} = 1.499$ vs. the archived 1.497—the nonadiabatic physics is reproduced quantitatively, not just the endpoint. Our adaptive growth realizes the same accuracy with moderately fewer gates than the archived runs (e.g. 592 vs. 780 CNOTs at $U = 10$; 120 vs. 352 at $U = 0$): adaptive-ansatz resources are tie-break-sensitive, and our leaner circuits make the noise thresholds reported below, if anything, slightly optimistic for the method. The gauge-dependent split between φ_{qc} and $\varphi_{G,2}$ differs from the archive (as it must between ansatz realizations); their gauge-invariant sum agrees.

We choose $T = 20$ for all four configurations: it is the shortest published loop time that is accurate across both regimes, it maximizes the interesting nonadiabatic stress on the cancellation mechanism (mid-loop infidelity up to 0.334 at $U = 10$), and—because $\varphi_{G,1}$ noise grows with T (Sec. 6)—it is also the most noise-favorable operating point, making our thresholds upper bounds within the published T range.

5 Noise thresholds

5.1 Threshold versus incoherent noise

Figure 1 shows the central result and Table 2 the extracted numbers. The error curves rise smoothly and steeply once the deterministic channel bias overtakes the sampling floor, and every configuration crosses the tolerance within the swept range. Past the crossing the curves are non-monotonic—the drifting phase wraps the circle, so its folded distance to the target oscillates and occasionally revisits small values (the $U = 10$ dips near $s = 0.2$)—which is why the pre-registered rule thresholds on the running-maximum envelope of the first crossing: accidental post-threshold revivals cannot inflate the reported budget. At $S_{\text{ref}} = 4096$ shots the topological ($\varphi_B = \pi$) configurations give $s^* = 0.28$ [0.27, 0.29] at $U = 0$ and $s^* = 0.0094$ [0.009, 0.0099] at $U = 10$; the trivial configurations give 0.52 and 0.0044. In device units the conservative per-interaction thresholds—the levels below which both phases of the same U are simultaneously accurate, hence distinguishable—are

$$U = 0: p_{2q} \lesssim 8.4 \times 10^{-4}, \quad U = 10: p_{2q} \lesssim 1.3 \times 10^{-5},$$

Table 2: Main results at $S_{\text{ref}} = 4096$ shots per circuit. For each configuration: the noiseless anchor phase and its deviation from the exact-diagonalization value, and the phase at the largest swept noise level below threshold (circular seed mean of 16 seeds, with the seed-mean circular deviation). The threshold column gives the incoherent-noise threshold in device units (equivalent two-qubit error at the trapped-ion profile ratios; brackets are 16/84 bootstrap CIs) and the shot floor S^* at $s = 0$. Depth/ N_{CX} : noiseless rows list the loop ansatz (layers of parameterized unitaries / loop CNOTs); rows marked (r) list the compiled interferometric readout circuit (scheduled layers / CNOTs including preparation, controlled gadgets and inverse preparation). Per-interaction conservative thresholds (worse of the two regimes): $s_{U=0}^* = 0.28$, $s_{U=10}^* = 0.0044$.

Configuration	Noise setting	φ_B	$ \varphi_B - \varphi_B^{\text{ED}} $	Threshold (device units)	Depth	N_{CX}
$U=0$ topological	noiseless	-3.1416	3.5×10^{-5}	$p_{2q} \leq 8.4 [8.2, 8.6] \times 10^{-4}$	15	120
	$s = 0.2 (6.0 \times 10^{-4})$	-2.851	0.290	$S^* = 16 [16, 22]$	243 ^(r)	198 ^(r)
$U=0$ trivial	noiseless	-0.0041	4.1×10^{-3}	$p_{2q} \leq 1.6 [1.5, 1.6] \times 10^{-3}$	12	124
	$s = 0.5 (1.5 \times 10^{-3})$	+0.377	0.377	$S^* = 16 [16, 16]$	224 ^(r)	196 ^(r)
$U=10$ topological	noiseless	-3.1415	7.3×10^{-5}	$p_{2q} \leq 2.8 [2.7, 3.0] \times 10^{-5}$	61	592
	$s = 0.007 (2.1 \times 10^{-5})$	-2.865	0.277	$S^* = 140 [116, 159]$	1086 ^(r)	1074 ^(r)
$U=10$ trivial	noiseless	-0.0000	1.7×10^{-5}	$p_{2q} \leq 1.3 [1.2, 1.4] \times 10^{-5}$	61	592
	$s = 0.003 (9.0 \times 10^{-6})$	+0.283	0.283	$S^* = 234 [181, 302]$	1135 ^(r)	1148 ^(r)

equivalently readout error $\lesssim 8.4 \times 10^{-4} / 2.8 \times 10^{-5}$ and $T_2 \gtrsim 3.6 \text{ s} / 110 \text{ s}$ at the profile ratios. The noninteracting thresholds sit within reach of the best reported trapped-ion two-qubit fidelities; the interacting ones are one to two orders of magnitude beyond current hardware. Below threshold the two regimes are cleanly separated: at every swept $s \leq s^*$, all 16 seeds of both regimes land within $\pi/2$ of their own quantized target (no misclassification events; logged discrimination checks), so the phase diagram read off the noisy estimates is correct with margin.

Two features of Table 2 demand explanation, and get one in Sec. 6. First, the U -dependence is enormous: the $64.2\times$ threshold gap between $U = 0$ and $U = 10$ far exceeds the $\sim 5\times$ ratio of their circuit sizes—naïve per-gate error accounting fails here, because the dynamical-phase leg multiplies the circuit-attenuation error by the energy scale $|E - \hbar|$ (≈ 12 at $U = 10$ vs. ≈ 4 at $U = 0$) and by the loop time. Second, the regime ordering flips with interaction: at $U = 0$ the trivial configuration is the more robust ($s^* = 0.52$ vs. 0.28), while at $U = 10$ it is the more fragile ($s^* = 0.0044$ vs. 0.0094)—at $U = 10$ the geometric leg’s bias changes sign and *adds* to the dynamical-phase drift instead of partially cancelling it.

5.2 The irreducible sampling floor

At $s = 0$ the only noise is finite sampling—the floor that survives even fault-tolerant hardware. The deviation scales as $S^{-1/2}$ [Fig. 2(a)] and crosses the tolerance at $S^* = 16 [16, 22]$ shots per circuit for the $U = 0$ topological configuration (the trivial one is within tolerance already at the smallest swept $S = 16$) and at $S^* = 140 [116, 159] / 234 [181, 302]$ for the $U = 10$ regimes (Table 2). Tens to a few hundred shots per circuit therefore suffice, but the *number of circuits* is large: the φ_G legs dominate the budget ($\sim 10^5$ circuit executions per estimate, Sec. 9), which is the practical cost of whole-pipeline phase reconstruction.

5.3 State damage versus phase damage

Comparing the phase error to the end-of-loop state infidelity under the same noise [Fig. 2(b)] reveals that the noiseless folklore—the topological phase is more robust than the state carrying it—survives only for the shallow circuits, and inverts for the deep ones. At $U = 0$ the phase stays within tolerance up to a loop-state infidelity of ~ 0.11 : the phase outlives the state, the noisy analogue of the method’s robustness to non-adiabaticity. At $U = 10$ the phase fails when the state is still at infidelity ~ 0.02 —the phase dies *first*, because the dynamical-phase reconstruction converts a percent-level energy attenuation into a tolerance-level phase drift through the $T|E - \hbar|$ lever arm. Robustness of the phase observable is therefore not intrinsic to the method; it is a competition between the bias-immune interferometry and the bias-amplifying reconstruction, and which side wins depends on the circuit depth and energy scale.

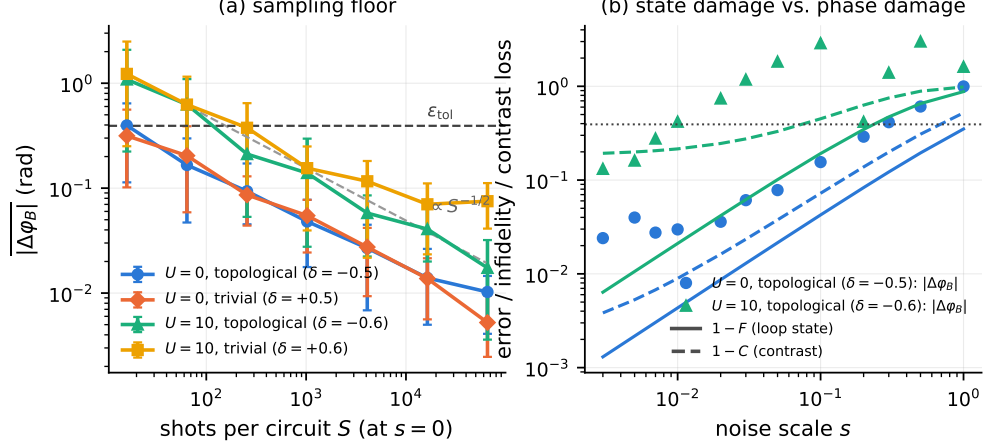


Figure 2: (a) Sampling floor: seed-mean circular Berry-phase deviation versus shots per circuit at $s = 0$ (24 seeds; guide line $\propto S^{-1/2}$); crossings of ϵ_{tol} define the shot floors S^* . (b) State damage versus phase damage: end-of-loop state infidelity $1 - F$ and interferometric contrast loss $1 - C$ of the noisy pipeline (exact channel values, lines) compared to the phase error (markers, from Fig. 1) for the topological configurations. At $U = 0$ (blue) the phase error stays below the state infidelity — the phase outlives the state — while at $U = 10$ (green) the ordering inverts: the reconstructed phase fails while the state is still at $\sim 98\%$ fidelity.

6 Why it breaks: error anatomy

The phase anatomy of Eq. (2) lets us assign every bit of damage to a leg, a channel, and a mechanism (Fig. 3). Throughout, “bias” is the deterministic shift of the infinite-shot noisy estimator and “fluctuation” the finite-shot seed spread; the two are logged separately for every leg and configuration.

The interferometric leg is structurally protected. Depolarizing-type noise on the Hadamard-test circuit shrinks both quadratures of $\langle W \rangle$ by a common contrast factor $C(s)$, so $\text{atan2}(\widehat{\text{Im}}, \widehat{\text{Re}})$ is (to first order) *unbiased*; the damage appears only as variance $\sim 1/(C\sqrt{S})^2$ per quadrature. This is visible in Fig. 3(a): the φ_{qc} bias stays at the 10^{-2} -rad level deep into the sweep (residual bias comes from the non-unit T_1 channel and asymmetries, not from depolarization), while its fluctuation blows up only when $C(s)$ approaches the sampling noise, $C \sim S^{-1/2}$. The two-quadrature estimator is essential here: a single-quadrature readout at a quantized target has zero slope and would convert contrast loss directly into bias.

The dynamical-phase leg is the weak point. The energy integrand of $\varphi_{G,1}$ is measured on circuits whose depth *grows* along the loop (the ansatz expands mostly in the first half), so channel attenuation pulls $\langle \hat{H} \rangle$ toward the maximally-mixed value $\bar{h} = \text{Tr} \hat{\mathcal{H}}/2^{N_q}$ by a λ -dependent factor. The time-reversed protocol cancels $\int \sigma \langle \hat{H} \rangle ds$ only if the attenuation is symmetric between the legs; it is not [Fig. 3(b)], and the residual is amplified by the loop time:

$$\delta\varphi_{G,1} \approx \frac{T}{2} \overline{[f_{\text{ret}}(\lambda) - f_{\text{fwd}}(\lambda)] (E(\lambda) - \bar{h})} \propto T \Delta N_{\text{CX}} s p_{2q}^{(0)}, \quad (4)$$

with ΔN_{CX} the forward/return circuit-size asymmetry and f the attenuation factors. This drift dominates the total bias for every configuration [Fig. 3(a)] and sets the threshold scale. Its testable content is the scaling: the measured $\varphi_{G,1}$ bias is linear in s over more than a decade [Fig. 3(a)], the measured attenuation asymmetry is concentrated exactly where the forward/return circuit sizes differ [Fig. 3(b)], and the leg biases of the four configurations order by their circuit-volume asymmetry (appendix Table 4). The noiseless dynamical-phase cancellation—exact under time reversal—is therefore the first casualty of hardware noise, and the mechanism is *structural*: any noise that attenuates expectation values asymmetrically between the two halves of the loop produces it. The geometric leg $\varphi_{G,2}$ is partially protected for the opposite reason: its integrand is weighted by the

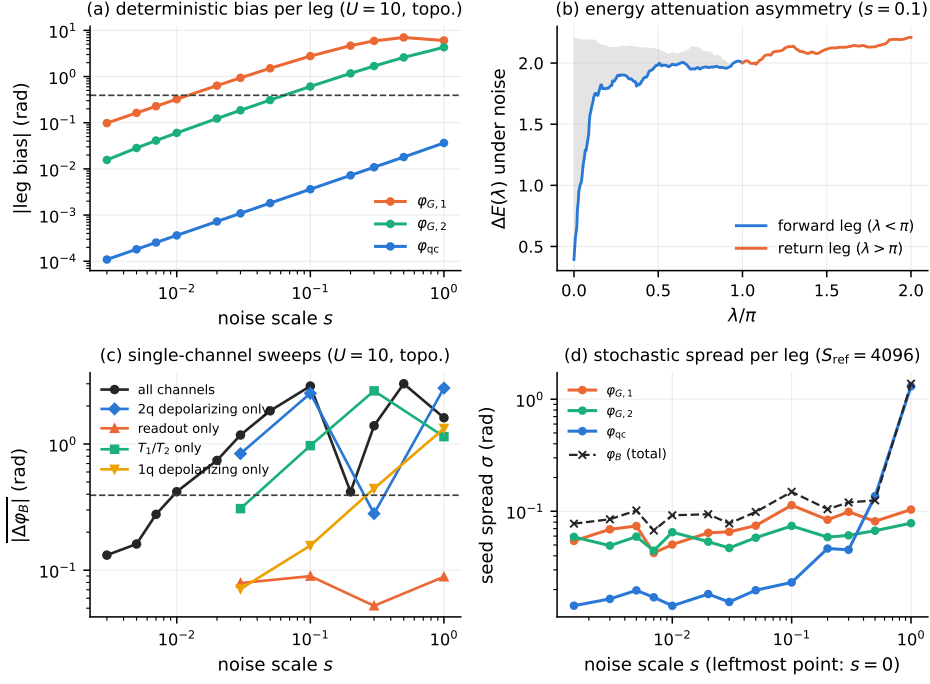


Figure 3: Error anatomy. (a) Deterministic bias of each phase leg (infinite-shot limit of the noisy estimator minus its noiseless value) versus noise scale, $U = 10$ topological configuration: the dynamical leg $\varphi_{G,1}$ dominates and grows linearly, the interferometric φ_{qc} is bias-free until its contrast collapses, and $\varphi_{G,2}$ is intermediate. (b) Measured energy shift $\Delta E(\lambda)$ at $s = 0.1$ on the forward and return legs: the return half is attenuated more (deeper circuits after mid-loop growth), so the time-reversal cancellation of $\int \langle \hat{\mathcal{H}} \rangle$ breaks by the shaded asymmetry. (c) Single-channel sweeps (each channel alone, scaled from the base profile): for these deep circuits two-qubit depolarizing noise and T_1/T_2 relaxation are comparable, one-qubit noise is much weaker, and readout error alone never crosses tolerance. (d) Stochastic seed spread of each leg at S_{ref} : variances of the independent legs add; the φ_G legs carry the largest share.

parameter increments $d\theta_\mu$, which are large only in the early, shallow part of the loop where attenuation is weakest.

Stochastic errors add; deterministic cancellations are accidental. The three legs are measured on separate circuits, so their shot fluctuations are independent: the measured seed variance of $\hat{\varphi}_B$ equals the sum of the leg variances within statistical accuracy [Fig. 3(d); appendix Table 4]. The co-evolution that cancels *deterministic* non-adiabatic residuals between φ_{qc} and $\varphi_{G,2}$ in the noiseless method buys nothing against sampling noise—there is no correlation to exploit. Whether the *deterministic* channel biases of the two φ_G legs cancel turns out to be configuration accident, not structure: at $U = 0$ they oppose (at $s = 0.1$, $\delta\varphi_{G,1} = 0.251$ against $\delta\varphi_{G,2} = -0.099$, a $\sim 40\%$ relief), while at $U = 10$ they *add* (2.773 and 0.609 in the topological regime; 3.268 and 3.736 in the trivial regime, where the geometric leg is the larger of the two). This sign flip is what makes the $U = 10$ trivial configuration the most fragile of the four and inverts the regime ordering between the interaction strengths. Nothing in the method protects these combinations; once noise attenuates the measured integrands, each leg drifts by its own circuit-weighted amount.

Which channel, which circuit property. Single-channel sweeps [Fig. 3(c)] rank the channels at the trapped-ion profile ratios. For the shallow $U = 0$ circuits, two-qubit depolarizing noise alone essentially reproduces the full threshold (single-channel $s^* = 0.35$ vs. 0.28 with all channels), and every other channel alone stays within tolerance across its entire sweep. For the deep $U = 10$ circuits, two-qubit depolarizing error and T_1/T_2 relaxation are *comparable* killers (single-channel thresholds < 0.03 and 0.035: the interacting measurement circuits run long enough that dephasing

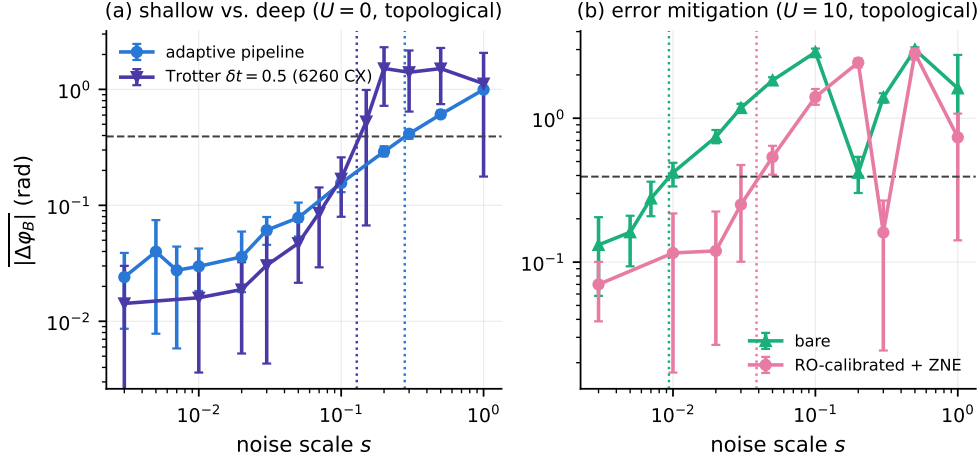


Figure 4: (a) Adaptive-variational versus Trotterized evolution under identical noise ($U = 0$ topological, $T = 20$, both at noiseless accuracy $\ll \epsilon_{\text{tol}}$): the 6260-CNOT Trotter loop crosses tolerance at $s^* = 0.13$, a factor ~ 2.2 below the 198-CNOT adaptive pipeline. (b) Mitigation: calibrated readout inversion plus CNOT-folded zero-noise extrapolation shifts the $U = 10$ topological error curve right by a factor ≈ 4.1 in s at $2\times$ circuit and calibration overhead.

across $\sim 10^3$ scheduled layers matters as much as gate error), one-qubit depolarizing noise is much weaker ($s^* = 0.25$), and readout error alone never crosses tolerance on our grid ($s^* > 1$)—readout bias attenuates parities by $(1 - 2\epsilon_{\text{ro}})^w$, a small effect at percent-level flip rates. The controlling combination is (circuit volume of the φ_G -integrand measurements) $\times$ (energy lever arm $T|E - \bar{h}|$), not gate count alone: it is why the $U = 10$ thresholds sit $64.2\times$ —not $\sim 5\times$ —below $U = 0$. Finally, a readout-only scope (sampling only the Hadamard test while treating φ_G as exact classical data) overstates the whole-method threshold by $\sim 3.6\times$ at $U = 0$ and by ~ 63.9 – $144.6\times$ at $U = 10$: a noise study of the interferometry alone would be off by up to two orders of magnitude for interacting systems.

7 Shallow versus deep: the compression dividend under noise

Folklore says shallow circuits should tolerate proportionally more noise. The measured answer is more interesting. We compare at matched noiseless accuracy on the $U = 0$ topological configuration: a first-order Trotterized time-reversed loop at $\delta t = 0.5$ —the coarsest step that retains noiseless accuracy ($|\Delta\varphi_B| = 2.7 \times 10^{-7}$)—costs 6260 CNOTs in its controlled interferometric form, against 198 for the adaptive readout circuit: a $\sim 30\times$ gate gap. Yet under the identical noise model, estimator, and S_{ref} , the Trotter baseline crosses tolerance at $s^* = 0.13$ [0.12, 0.14] (equivalent $p_{2q} \approx 3.9 \times 10^{-4}$; after the first sweep located the crossing between grid points 0.05 and 0.2 we added two refinement levels, $s = 0.07, 0.15$ —disclosed here, all points reported)—only a factor ~ 2.2 below the adaptive threshold [Fig. 4(a)]. The structural reason follows from Sec. 6: the Trotter interferometer needs no φ_G reconstruction (the controlled propagator carries its physical phase), so it inherits the full bias-immunity of the two-quadrature readout—its exact phase stays within 0.035 rad of the target even at $s = 0.1$ where its contrast has collapsed to $C = 0.069$ —and it fails only when contrast meets shot noise, $C(s) \sim S^{-1/2}$. The adaptive method keeps its contrast high ($C = 0.928$ at $s = 0.1$) but spends most of that advantage on the reconstruction bias of its φ_G legs. Compression, in other words, pays off fully only where the phase is measured interferometrically; the classical reconstruction squanders the depth advantage. Two corollaries follow. First, the dividend re-appears in the shot budget: at $s = 0.1$ the Trotter estimator needs $\sim (1/C^2) \approx 200\times$ more shots per quadrature than the adaptive one for equal phase variance, and past $s \approx 0.2$ (where $C < 10^{-2}$) no realistic budget rescues it, while the mitigated adaptive pipeline still resolves the phase at $s = 0.55$ (Sec. 8). Second, the comparison sharpens the design message: an adaptive method with a bias-defended (or interferometric) φ_G would enjoy the full $\sim 30\times$ compression dividend. At the published matched-accuracy step $\delta t = 0.005$ (6.24×10^5 CNOTs [6]) the circuit is beyond hope at any swept noise level: a direct density-matrix

run at our smallest scale $s = 0.003$ leaves an ancilla contrast of 4.5×10^{-4} —indistinguishable from zero at any realistic shot budget, confirming the analytic estimate $C \approx e^{-\lambda N_{\text{CX}}}$.

8 Error mitigation raises the threshold

Because the dominant failure is a deterministic attenuation bias, standard mitigation should help, and it does [Fig. 4(b)]. We compose (i) readout mitigation with flip rates *calibrated from sampled calibration circuits* (no oracle access to the true ϵ_{ro}), inverting the confusion model on every measured parity, and (ii) observable-level zero-noise extrapolation [20–22]: every CNOT is folded ($\text{CX} \rightarrow \text{CX}^3$), all observables are measured at both scale factors, and each is linearly extrapolated to zero noise before the phase is assembled. At $2\times$ circuit overhead plus calibration this shifts the $U = 10$ topological threshold from $s^* = 0.0094$ to 0.039 [$0.037, 0.04$]—a factor 4.1, bringing the equivalent two-qubit budget to 1.2×10^{-4} —and the $U = 0$ topological threshold from 0.28 to 0.55 [$0.54, 0.56$] (factor 2.0; here the bare threshold is already partly variance-limited, so removing bias buys less). Mitigation attacks exactly the leg our anatomy identified: extrapolation flattens the attenuation drift of the φ_G legs, while the already-unbiased interferometric leg gains little beyond its readout correction. The larger relative gain at $U = 10$ —precisely the regime where the bare method is weakest—supports the design message that defending the reconstruction, not the interferometry, is where mitigation effort pays.

9 Resource accounting

The classical costs of our study are modest: each adaptive trajectory integrates $\sim 2\text{--}6 \times 10^3$ RK4 steps in minutes on a workstation, and the full density-matrix campaign (96 probability caches spanning configurations, noise scales, channel isolations and fold factors, each simulating every measurement circuit at ~ 500 trajectory nodes) totals roughly fifty CPU-hours. The device-side accounting is the informative one: one whole-pipeline Berry-phase estimate at S_{ref} costs $2S_{\text{ref}}$ interferometric shots but $\sim 2.8 \times 10^8$ shots in total for $U = 10$ ($\sim 5.8 \times 10^7$ at $U = 0$), over $\sim 10^5$ distinct circuit executions, $>99\%$ of them φ_G -integrand measurements. On hardware these measurements are not extra work—the same expectation values are consumed by the McLachlan construction itself—but a Berry-phase experiment that could avoid reconstructing φ_G (e.g. by enforcing a phase-tracking ansatz) would collapse its sampling budget by two orders of magnitude. This, together with Eq. (4), is our main design message for hardware demonstrations.

10 Related work

Quantum algorithms for topological invariants span quench protocols, VQE-based wave-function topology, and adiabatic interferometry [4, 5, 24]; the adaptive variational Berry-phase method we characterize is Ref. [6], building on AVQDS [8], AVQITE [7], and TETRIS-style growth [9], with the variational-simulation theory of [15, 16]. Two recent lines are adjacent but complementary to ours. Zhang *et al.* [27] study *sampling* noise inside the variational imaginary-time *construction* loop (shot allocation for the M, V estimates in ground-state preparation, with Tikhonov stabilization)—the construction axis we deliberately freeze—whereas we propagate device noise and sampling through the entire Berry-phase *measurement* pipeline and extract topological-accuracy thresholds. And Ref. [28] cancels *deterministic* finite-runtime (adiabatic) errors in Hadamard-test Berry-phase estimation by runtime randomization in an early-fault-tolerant setting, orthogonal to the stochastic hardware-error axis characterized here. We are not aware of a published noise-resilience characterization of adaptive-variational geometric-phase estimation; the closest antecedents beyond the above are hardware Berry-phase interferometry on single- and few-qubit systems [4, 25] and general NISQ error analyses [19]. Our mitigation stack uses standard components [20–23], and our qubit-wise measurement grouping follows [26]. The failure mechanism we identify—time-reversal-asymmetric attenuation integrating into a dynamical-phase drift—does not, to our knowledge, appear in prior noise studies of variational simulation.

11 Limitations

Our thresholds characterize the measurement resilience of the method at its noiseless variational trajectory. Re-deriving the trajectory from noisy M and V (a stochastic McLachlan flow with noisy ansatz growth) is the natural next layer; we expect it to lower thresholds further, since M -inversion amplifies sampling noise, and our frozen-trajectory numbers should be read as upper bounds on full-pipeline resilience. Second, the four-channel profile co-varies along a single scalar at trapped-ion ratios; other platforms reweight the channels (our single-channel sweeps in Fig. 3(c) let a reader re-derive rough thresholds for other mixes), and our model omits coherent over-rotations and crosstalk. Third, the study is at $N = 4$ (8 qubits): the mechanism of Eq. (4) predicts thresholds falling with both circuit volume and loop time, so scaling to larger chains will tighten every number; the anatomy, not the absolute thresholds, is the transferable result. Finally, the quantitative thresholds inherit our leaner-than-archive adaptive circuits (Sec. 4); for the archived circuit sizes they would scale down by the corresponding CNOT ratios ($\sim 1.3\text{--}3\times$).

12 Conclusions

We have measured, under a realistic device model applied to every circuit of the pipeline, how much noise adaptive variational Berry-phase estimation actually tolerates. The verdict is two-sided. For the noninteracting chain the method is genuinely near-term: the topological phase survives to an equivalent two-qubit error of $\sim 8.4 \times 10^{-4}$ at 4096 shots per circuit, within about a factor of two of the best reported trapped-ion entangling fidelities, with an irreducible sampling floor of only tens of shots per circuit. For the interacting chain the threshold collapses to $\sim 1.3 \times 10^{-5}$ —a $64.2\times$ drop that outruns the $\sim 5\times$ growth in circuit size—because the failure is not where intuition puts it. The interferometric readout, whose fragility motivates most noise worries, is structurally bias-immune; what breaks is the classical reconstruction of the dynamical phase from measured energies, whose time-reversal cancellation is destroyed by depth-asymmetric attenuation and amplified by the loop time and the energy scale. The same anatomy explains why a $30\times$ deeper Trotter interferometer—bias-free by construction—loses to the adaptive method by only $\sim 2.2\times$ at equal shots; why the noiseless folklore that the phase outlives the state inverts for deep circuits; and why bias-targeted mitigation (readout calibration plus zero-noise extrapolation) buys its largest factor ($4.1\times$) exactly where the bare method is weakest. Defending the dynamical-phase leg—symmetrized scheduling, attenuation-aware rescaling, or ansätze that keep the phase interferometric—is, on this evidence, the highest-leverage step toward measuring many-body topological invariants on near-term hardware.

Acknowledgments and Disclosure of Funding

This study re-implements and characterizes the algorithm of Mootz and Yao [6]; we thank the authors for publishing complete data records, which made the independent cross-validation in Sec. 4 possible.

References

- [1] M. V. Berry. Quantal phase factors accompanying adiabatic changes. *Proc. R. Soc. Lond. A*, 392:45–57, 1984.
- [2] D. Xiao, M.-C. Chang, and Q. Niu. Berry phase effects on electronic properties. *Rev. Mod. Phys.*, 82:1959–2007, 2010.
- [3] J. Zak. Berry’s phase for energy bands in solids. *Phys. Rev. Lett.*, 62:2747–2750, 1989.
- [4] B. Murta, G. Catarina, and J. Fernández-Rossier. Berry phase estimation in gate-based adiabatic quantum simulation. *Phys. Rev. A*, 101:020302(R), 2020.
- [5] X. Xiao, J. K. Freericks, and A. F. Kemper. Robust measurement of wave function topology on NISQ quantum computers. *Quantum*, 7:987, 2023.
- [6] M. Mootz and Y.-X. Yao. Efficient Berry phase calculation via adaptive variational quantum computing approach. *APL Quantum*, 3:016111, 2026. arXiv:2506.19150.

- [7] N. Gomes, A. Mukherjee, F. Zhang, T. Iadecola, C.-Z. Wang, K.-M. Ho, P. P. Orth, and Y.-X. Yao. Adaptive variational quantum imaginary time evolution approach for ground state preparation. *Adv. Quantum Technol.*, 4:2100114, 2021.
- [8] Y.-X. Yao, N. Gomes, F. Zhang, C.-Z. Wang, K.-M. Ho, T. Iadecola, and P. P. Orth. Adaptive variational quantum dynamics simulations. *PRX Quantum*, 2:030307, 2021.
- [9] P. G. Anastasiou, Y. Chen, N. J. Mayhall, E. Barnes, and S. E. Economou. TETRIS-ADAPT-VQE: An adaptive algorithm that yields shallower, denser circuit ansätze. *Phys. Rev. Research*, 6:013254, 2024.
- [10] W. P. Su, J. R. Schrieffer, and A. J. Heeger. Solitons in polyacetylene. *Phys. Rev. Lett.*, 42:1698–1701, 1979.
- [11] S. R. Manmana, A. M. Essin, R. M. Noack, and V. Gurarie. Topological invariants and interacting one-dimensional fermionic systems. *Phys. Rev. B*, 86:205119, 2012.
- [12] P. Jordan and E. Wigner. Über das Paulische Äquivalenzverbot. *Z. Phys.*, 47:631–651, 1928.
- [13] R. D. King-Smith and D. Vanderbilt. Theory of polarization of crystalline solids. *Phys. Rev. B*, 47:1651–1654, 1993.
- [14] R. Resta. Macroscopic polarization in crystalline dielectrics: the geometric phase approach. *Rev. Mod. Phys.*, 66:899–915, 1994.
- [15] A. D. McLachlan. A variational solution of the time-dependent Schrödinger equation. *Mol. Phys.*, 8:39–44, 1964.
- [16] X. Yuan, S. Endo, Q. Zhao, Y. Li, and S. C. Benjamin. Theory of variational quantum simulation. *Quantum*, 3:191, 2019.
- [17] A. K. Ekert, C. M. Alves, D. K. L. Oi, M. Horodecki, P. Horodecki, and L. C. Kwek. Direct estimations of linear and nonlinear functionals of a quantum state. *Phys. Rev. Lett.*, 88:217901, 2002.
- [18] S. A. Moses *et al.* A race-track trapped-ion quantum processor. *Phys. Rev. X*, 13:041052, 2023.
- [19] J. Preskill. Quantum computing in the NISQ era and beyond. *Quantum*, 2:79, 2018.
- [20] K. Temme, S. Bravyi, and J. M. Gambetta. Error mitigation for short-depth quantum circuits. *Phys. Rev. Lett.*, 119:180509, 2017.
- [21] Y. Li and S. C. Benjamin. Efficient variational quantum simulator incorporating active error minimization. *Phys. Rev. X*, 7:021050, 2017.
- [22] A. Kandala, K. Temme, A. D. Córcoles, A. Mezzacapo, J. M. Chow, and J. M. Gambetta. Error mitigation extends the computational reach of a noisy quantum processor. *Nature*, 567:491–495, 2019.
- [23] S. Bravyi, S. Sheldon, A. Kandala, D. C. McKay, and J. M. Gambetta. Mitigating measurement errors in multiqubit experiments. *Phys. Rev. A*, 103:042605, 2021.
- [24] M. Atala, M. Aidelsburger, J. T. Barreiro, D. Abanin, T. Kitagawa, E. Demler, and I. Bloch. Direct measurement of the Zak phase in topological Bloch bands. *Nat. Phys.*, 9:795–800, 2013.
- [25] P. J. Leek *et al.* Observation of Berry’s phase in a solid-state qubit. *Science*, 318:1889–1892, 2007.
- [26] V. Verteletskyi, T.-C. Yen, and A. F. Izmaylov. Measurement optimization in the variational quantum eigensolver using a minimum clique cover. *J. Chem. Phys.*, 152:124114, 2020.
- [27] F. Zhang, N. Gomes, J. Aftergood, T. Iadecola, Y.-X. Yao, and P. P. Orth. Sampling noise and optimized measurement distribution in imaginary-time quantum dynamics simulations. arXiv:2605.20378, 2026.
- [28] C. Kiumi. Adiabatic error cancellation in Berry phase estimation. arXiv:2604.20952, 2026.

A Extended data and implementation notes

A.1 All swept points

Table 3: Complete sweep record. Left: every swept incoherent-noise level s : seed mean $\pm$ seed standard deviation of the circular Berry-phase deviation $|\Delta\varphi_B|$ (rad) at $S_{\text{ref}} = 4096$, 16 seeds per point. Right block (lower): shots axis at $s = 0$ (24 seeds). No level or seed is omitted ($s = 0.005, 0.007$ are the disclosed post-first-pass refinement levels, swept for all four configurations).

s	$U=0$ topological	$U=0$ trivial	$U=10$ topological	$U=10$ trivial
0	0.023 ± 0.013	0.023 ± 0.018	0.047 ± 0.030	0.107 ± 0.073
0.003	0.024 ± 0.015	0.029 ± 0.020	0.132 ± 0.069	0.283 ± 0.109
0.005	0.040 ± 0.029	0.024 ± 0.019	0.161 ± 0.081	0.433 ± 0.105
0.007	0.028 ± 0.021	0.030 ± 0.018	0.277 ± 0.067	0.608 ± 0.139
0.01	0.030 ± 0.016	0.031 ± 0.015	0.419 ± 0.092	0.812 ± 0.172
0.02	0.036 ± 0.022	0.023 ± 0.013	0.741 ± 0.094	1.541 ± 0.114
0.03	0.061 ± 0.020	0.037 ± 0.022	1.178 ± 0.078	2.381 ± 0.107
0.05	0.078 ± 0.026	0.047 ± 0.030	1.831 ± 0.099	2.530 ± 0.143
0.1	0.156 ± 0.023	0.096 ± 0.027	2.877 ± 0.147	0.737 ± 0.135
0.2	0.290 ± 0.032	0.160 ± 0.030	0.418 ± 0.104	0.131 ± 0.108
0.3	0.414 ± 0.031	0.240 ± 0.045	1.395 ± 0.120	2.111 ± 0.120
0.5	0.608 ± 0.039	0.377 ± 0.036	3.001 ± 0.110	2.525 ± 0.181
1	0.996 ± 0.038	0.682 ± 0.048	1.613 ± 1.000	1.751 ± 0.912

S (at $s = 0$)	$U=0$ topological	$U=0$ trivial	$U=10$ topological	$U=10$ trivial
16	0.396	0.315	1.086	1.227
64	0.166	0.204	0.627	0.626
256	0.093	0.086	0.211	0.376
1024	0.048	0.055	0.139	0.155
4096	0.027	0.027	0.058	0.116
16384	0.014	0.014	0.041	0.070
65536	0.010	0.005	0.017	0.075

A.2 Anchor cross-validation traces

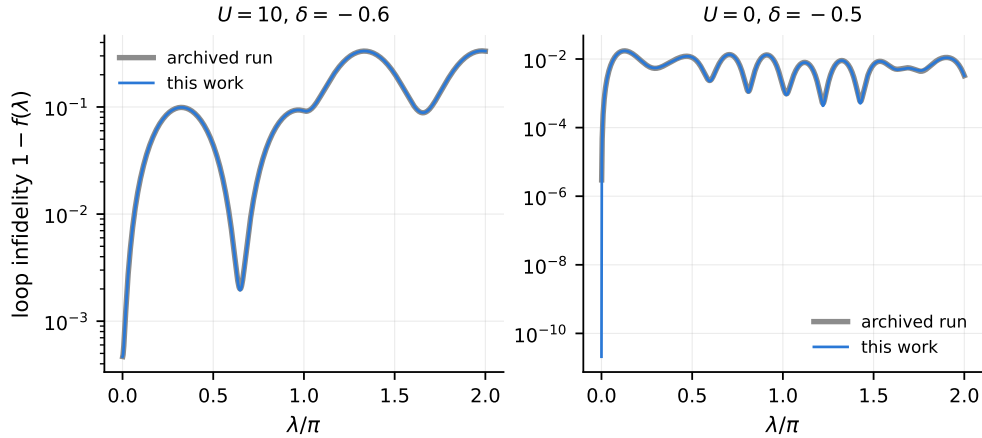


Figure 5: Per- λ loop infidelity $1 - f(\lambda)$ of our noiseless re-implementation (thin blue) overlaid on the corresponding archived runs of Ref. [6] (thick gray) at the same $(\delta, T=20)$: the nonadiabatic oscillation pattern, envelope, and peak values agree quantitatively (cf. Table 1), evidencing that the re-implemented variational dynamics—not merely the endpoint phase—matches the published method.

Table 4: Per-leg error anatomy at two representative noise scales. Bias: infinite-shot limit of the noisy estimator minus the noiseless leg value. Spread: seed standard deviation at $S_{\text{ref}} = 4096$ (circular for φ_{qc}). Last column: measured variance of $\hat{\varphi}_B$ vs. the sum of leg variances (independence check).

Configuration	scale	bias(φ_{qc})	bias($\varphi_{G,1}$)	bias($\varphi_{G,2}$)	$\sigma(\varphi_{\text{qc}})$	$\sigma(\varphi_{G,1})$	$\sigma(\varphi_{G,2})$	$\text{Var}[\varphi_B] / \sum_{\ell} \text{Var}[\varphi_{\ell}]$
$U=0$ topological	s=0.1	-0.000	+0.251	-0.099	0.014	0.012	0.012	0.0005 / 0.0005
$U=0$ topological	s=1	-0.004	+1.902	-0.882	0.024	0.018	0.021	0.0014 / 0.0013
$U=0$ trivial	s=0.1	-0.001	+0.149	-0.056	0.012	0.015	0.019	0.0007 / 0.0007
$U=0$ trivial	s=1	-0.009	+1.181	-0.486	0.031	0.033	0.019	0.0023 / 0.0024
$U=10$ topological	s=0.1	+0.004	+2.773	+0.609	0.023	0.113	0.074	0.0222 / 0.0188
$U=10$ topological	s=1	+0.036	+6.038	+4.305	1.304	0.104	0.078	1.8889 / 1.7170
$U=10$ trivial	s=0.1	+0.000	+3.268	+3.736	0.019	0.060	0.095	0.0183 / 0.0130
$U=10$ trivial	s=1	+0.002	+8.069	+20.785	1.521	0.093	0.109	2.5897 / 2.3343

A.3 Mechanism numbers

A.4 Implementation details

Software. All components (symplectic Pauli algebra, Jordan–Wigner construction, exact diagonalization, the AVQITE/AVQDS engine, the circuit compiler, the density-matrix simulator with noise channels, and the analysis) were implemented for this study in Python/NumPy; no quantum-simulation framework is used. The Hamiltonian construction is verified against an independent second-quantized diagonalization with explicit fermionic signs, and the compiled circuits against dense matrix references ($\exp(-i\theta P)$ exactly reproduced for representative strings up to weight 7; controlled rotations block-diagonal to 10^{-12}). The noise channels are unit-tested against analytic depolarizing contrast factors, e^{-t/T_1} population decay, e^{-t/T_2} coherence decay, and readout confusion tables. All run configurations, per-step trajectories, probability caches, sampled estimates, and analysis outputs are logged; every number in this paper is generated from those logs at build time.

Jordan–Wigner layout and reference states. Qubit $2j$ hosts the spin-down and qubit $2j+1$ the spin-up (twisted) mode of site j ; the twist acts on the odd-qubit boundary string. At half filling the reference product state is spin-up on sites 1, 2 and spin-down on sites 3, 4 for $\delta < 0$; for $\delta > 0$ the imaginary-time flow from that state stalls on an excited stationary point of the variational manifold (all pool gradients vanish while $\text{var}[\hat{\mathcal{H}}]$ is still large), and the doubly-occupied dimer-site reference (both species on sites 1 and 4) is used instead, matching the reference-state switch in the authors’ data records. With it, AVQITE converges to the true ground state in every configuration (Table 1).

McLachlan solve. The linear systems $M\dot{\theta} = \sigma V$ (and their growth-candidate extensions) are solved with Tikhonov regularization $\varepsilon = 10^{-7}$. The regularization is required for a measurable protocol, not for accuracy: with a minimal-norm pseudo-inverse the component $\dot{\theta}_{\mu}(t)$ jumps between near-degenerate ansatz directions from step to step (the state evolution and all gauge-invariant phases are unaffected), and the $\varphi_{G,2}$ integrand inherits those jumps, so no finite measurement cadence could integrate it. With Tikhonov damping the velocities are continuous, the integrand is smooth between growth events, and the node-based estimator of Eq. (3) converges; anchor observables change at the 10^{-4} level or below.

Quadrature nodes. Nodes are snapped to integrator steps, approximately uniformly in wall-clock time, with the leg boundary and endpoints always included (482 nodes per configuration). The estimator interpolates measured observables linearly between nodes and integrates them against the exact stored parameter increments; its zero-noise infinite-shot bias against the anchor is 9.1×10^{-3} ($U=0$ topological), 2.7×10^{-3} ($U=0$ trivial), 3.5×10^{-3} ($U=10$ topological), and 0.0641 ($U=10$ trivial) rad — 1–10% of the tolerance and included in every reported deviation (no correction is applied).

Energy measurement. The Hamiltonian strings are grouped greedily into qubit-wise commuting families (10, 12, 13 and 11 groups for the $U=0$ topological/trivial and $U=10$ topological/trivial configurations respectively; the greedy grouping order depends on the coefficient magnitudes, hence on δ); each family is measured by one basis rotation and a computational-basis readout of all 8 system qubits, all strings of a family being estimated from the same S samples (their covariance is therefore

physical). The identity component ($-U$ at interaction U) is a classical constant. Readout confusion acts on every measured qubit; the parity of a weight- w string is attenuated by $(1 - 2\epsilon_{\text{ro}})^w$.

Reproducibility of sampling. Every sampled estimate is seeded as `crc32(tag|s|S|seed)`, so any single number in the results files can be regenerated in isolation.

Berry-phase branch handling. $\varphi_{G,1}, \varphi_{G,2}$ accumulate unwrapped; $\hat{\varphi}_{\text{qc}} = \text{atan2}$ lies in $(-\pi, \pi]$; the assembled $\hat{\varphi}_B$ is folded once to $(-\pi, \pi]$ and compared to the reference by circular distance. The $U=10$ raw phases sit at odd multiples of π (e.g. -9.425 at $T=20$), i.e. exactly on the fold boundary; circular comparison makes the reported deviations insensitive to which side of $\pm\pi$ a noisy estimate lands on.